PROFESSIONAL INSTALLATION MANUAL

Ubuntu Server LTS Deployment for ABC Startup Solutions

ITEP 414 – System Administration and Maintenance | Week 3

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1. Introduction

This manual provides a repeatable deployment procedure for an Ubuntu Server LTS virtual machine. It emphasizes consistency, verification, security, and documentation.

2. **Deployment Standards**

Standard	**Target**
VM Name	Ubuntu-Server-Week03
Hostname	server01
RAM / CPU	4 GB / 2 vCPU
Storage	40 GB VDI/VMDK
Network	NAT or Bridged when required
Administration	Non-root sudo user + SSH
Updates	apt update and apt upgrade after installation

3. **Ubuntu Server Installation**

Create the VM, attach the Ubuntu Server LTS ISO, boot the installer, select English, configure the keyboard, accept DHCP, set hostname to server01, create a non-root administrator, select guided full-disk installation, enable OpenSSH Server, and complete the installation.

Acceptance criteria: the VM boots from the installed disk; hostname is server01; the administrator can log in; an IP address is assigned; SSH is running; and updates complete without blocking errors.

4. **Initial Configuration**

hostname

ip addr

sudo apt update

sudo apt upgrade -y

systemctl status ssh

Record the IP address and installation date. Do not publish passwords, tokens, or private keys

5. Acceptance Testing

Test	**Command / Action**	**Pass Criteria**
Login	Console login	Successful authentication
Hostname	hostname	server01
Network	ip addr	Valid IP
Internet	ping -c 4 google.com	Replies received
Updates	apt update / upgrade	No blocking errors
SSH	systemctl status ssh	active (running)

6. Troubleshooting

For network problems, check the VM adapter, DHCP, interface state, routing, and DNS. For SSH problems, verify the package and service. For update failures, verify connectivity and package-manager errors. For boot failures, verify the virtual disk and boot order.

7. Security and Best Practices

Use least privilege, strong passwords, regular updates, SSH access controls, firewall rules, backups, and documented changes. Avoid exposing SSH directly to the public internet without a controlled security design.

8. Windows Server Evaluation Summary

Create a separate VM, attach the Windows Server Evaluation ISO, install the OS, assign a computer name, create an administrator password, log in, and capture installation evidence. Record the exact Windows Server version used.

9. Operating System Comparison

Area	**Windows Server**	**Ubuntu Server**	**Rocky Linux**
Licensing	Commercial/evaluation	Open source	Open source
Administration	GUI + PowerShell	CLI + tools	CLI + tools
Packages	Windows Update/MSI	APT	DNF
Security	Microsoft ecosystem	Linux permissions/AppArmor/tools	SELinux/Linux tools
Typical Use	AD and Microsoft workloads	Web, cloud, development, services	Enterprise Linux/RHEL-compatible
Strength	Microsoft integration	Large community	Enterprise compatibility
Limitation	Licensing/resource needs	Linux learning curve	More Linux administration knowledge

10. Handover Checklist

☐ VM specifications recorded

☐ Hostname and IP recorded

☐ Screenshots stored

☐ SSH verified

☐ Updates completed

☐ BIOS/UEFI comparison attached

☐ Boot flowchart attached

☐ Troubleshooting notes recorded

☐ No credentials committed to GitHub

11. References

Ubuntu Documentation – https://documentation.ubuntu.com/

Microsoft Learn – https://learn.microsoft.com/

Rocky Linux Documentation – https://docs.rockylinux.org/

VirtualBox – https://www.virtualbox.org/wiki/Documentation

UEFI Forum – https://uefi.org/